

1. (1.5 marks) A security code consists of one letter followed by two digits. If the letter is already fixed as A and digits may repeat, how many different codes are possible?

- a) 10
- b) 20
- c) 100
- d) 200
- e) 1,000

2. (1.5 marks) In a class of 80 students, 46 use the bus, 32 use carpooling, and 14 use both. What is the probability that a randomly selected student uses the bus or carpooling?

- a) 0.625
- b) 0.700
- c) 0.750
- d) 0.800
- e) 0.925

3. (1.5 marks) Each manufactured component is defective with probability 0.10, independently of the others. What is the probability that at least one of 3 components is defective?

- a) 0.100
- b) 0.271
- c) 0.300
- d) 0.729
- e) 0.900

4. (1.5 marks) Of 60 trainees, 24 attended a workshop and 12 of those who attended received an excellent rating. If one workshop attendee is selected at random, what is the probability that the trainee received an excellent rating?

- a) 0.20
- b) 0.30
- c) 0.40
- d) 0.50
- e) 0.60

5. (1.5 marks) Events A and B are independent with $P(A) = 0.50$ and $P(B) = 0.40$. Find $P(A \cap \bar{B})$.

- a) 0.10
- b) 0.20
- c) 0.30
- d) 0.40
- e) 0.50

6. (1.5 marks) A warehouse receives 60% of its parts from Supplier A and 40% from Supplier B. The defective rates are 2% and 5%, respectively. If a randomly selected part is defective, what is the probability that it came from Supplier A?

- a) 0.250
- b) 0.375
- c) 0.500
- d) 0.625
- e) 0.750

7. (1.5 marks) A random variable X has the distribution:

$$f(x) = \begin{cases} k & x = 0 \\ 2k & x = 1 \\ 3k & x = 2 \\ 4k & x = 3 \end{cases}$$

What is k ?

- a) 0.05
- b) 0.10
- c) 0.15
- d) 0.20
- e) 0.25

8. (1.5 marks) The random variable X has values 0, 1, and 2 with probabilities 0.20, 0.50, and 0.30, respectively. Find $E[X]$.

- a) 0.8
- b) 1.0
- c) 1.1
- d) 1.3
- e) 1.5

9. (1.5 marks) A random variable X has values -1 , 0, and 1 with probabilities 0.25, 0.50, and 0.25, respectively. Find $Var(X)$.

- a) 0.25
- b) 0.50
- c) 0.75
- d) 1.00
- e) 1.50

10. (1.5 marks) If $E[X] = 3$ and $Y = 5X + 2$, find $E[Y]$.

- a) 10
- b) 15
- c) 17
- d) 18
- e) 25

11. (1.5 marks) If the standard deviation of X is 2 and $Y = 4X - 1$, what is the standard deviation of Y ?

- a) 2
- b) 4
- c) 7
- d) 8
- e) 15

12. (1.5 marks) For a discrete random variable X :

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.25 & 1 \leq x < 2 \\ 0.65 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Find $P(X = 2)$.

- a) 0.20
- b) 0.25
- c) 0.40
- d) 0.65
- e) 0.90

13. (1.5 marks) An assembly line records the number of scratched items among 20 independently produced items, where each item has the same probability of being scratched. Which distribution models the number of scratched items?

- a) Binomial
- b) Poisson

- c) Geometric
- d) Hypergeometric
- e) Exponential

14. (1.5 marks) A quality inspector checks 5 products independently, where each product is defective with probability 0.20. What is the probability of finding exactly one defective product?

- a) 0.2048
- b) 0.3277
- c) 0.4096
- d) 0.5904
- e) 0.6723

15. (1.5 marks) A printer fails on any job with probability 0.10, independently. What is the probability of at least one failure in 4 print jobs?

- a) 0.2439
- b) 0.3439
- c) 0.4096
- d) 0.6561
- e) 0.9000

16. (1.5 marks) The number of calls arriving at a help desk in one minute follows a Poisson distribution with mean 2. What is the probability of receiving no calls in a minute?

- a) 0.0498
- b) 0.1353
- c) 0.2707
- d) 0.4060
- e) 0.8647

17. (1.5 marks) The number of faults on a sheet follows a Poisson distribution with mean 3. What is the probability that a sheet has at most one fault?

- a) 0.0498
- b) 0.1494
- c) 0.1991
- d) 0.4232
- e) 0.8009

18. (1.5 marks) A machine produces an acceptable item with probability 0.25 on each independent trial. What is the probability that the first acceptable item occurs on the 3rd trial?

- a) 0.0469
- b) 0.1055
- c) 0.1406
- d) 0.1875
- e) 0.4219

19. (1.5 marks) A salesperson makes a sale with probability 0.50 on each independent call. What is the probability that the 2nd sale occurs on the 4th call?

- a) 0.0625
- b) 0.1250
- c) 0.1875
- d) 0.2500
- e) 0.3750

20. (1.5 marks) A box contains 40 batteries, of which 10 are defective. If 8 batteries are selected without replacement, what is the expected number of defective batteries selected?

- a) 1

- b) 2
- c) 2.5
- d) 4
- e) 8

21. (1.5 marks) A waiting time X is uniformly distributed from 0 to 10 minutes. Find $P(2 < X < 5)$.

- a) 0.20
- b) 0.30
- c) 0.40
- d) 0.50
- e) 0.70

22. (1.5 marks) A delivery time X is uniformly distributed from 0 to 20 minutes. Find the time t such that $P(X \leq t) = 0.75$.

- a) 5 minutes
- b) 10 minutes
- c) 12 minutes
- d) 15 minutes
- e) 18 minutes

23. (1.5 marks) The lifetime of a sensor follows an exponential distribution with mean 5 years. What is the probability that it lasts more than 10 years?

- a) 0.1353
- b) 0.2707
- c) 0.3679
- d) 0.6321
- e) 0.8647

24. (1.5 marks) The time until the next customer arrives is exponentially distributed with rate 0.5 per minute. Find $P(X < 4)$.

- a) 0.1353
- b) 0.3679
- c) 0.6321
- d) 0.8647
- e) 0.9817

25. (1.5 marks) Test scores are normally distributed with mean 60 and standard deviation 10. What is $P(X < 70)$?

- a) 0.1587
- b) 0.3413
- c) 0.5000
- d) 0.6826
- e) 0.8413

26. (1.5 marks) Package weights are normally distributed with mean 100 g and standard deviation 15 g. What is $P(85 < X < 115)$?

- a) 0.1587
- b) 0.3174
- c) 0.5000
- d) 0.6826
- e) 0.8413

27. (1.5 marks) A continuous random variable X has probability density function $f(x) = cx$ for $0 < x < 2$, and 0 otherwise. Find c .

- a) 0.25

- b) 0.50
- c) 1.00
- d) 1.50
- e) 2.00

28. (1.5 marks) A continuous random variable X has density $f(x) = 2x$ for $0 < x < 1$, and 0 otherwise. Find $P(X > 0.5)$.

- a) 0.25
- b) 0.50
- c) 0.625
- d) 0.75
- e) 0.875

Q	Ans.	Topic	Q	Ans.	Topic
1	c	Probability	15	b	Discrete Distributions
2	d	Probability	16	b	Discrete Distributions
3	b	Probability	17	c	Discrete Distributions
4	d	Probability	18	c	Discrete Distributions
5	c	Probability	19	c	Discrete Distributions
6	b	Probability	20	b	Discrete Distributions
7	b	Random Variables	21	b	Continuous Distributions
8	c	Random Variables	22	d	Continuous Distributions
9	b	Random Variables	23	a	Continuous Distributions
10	c	Random Variables	24	d	Continuous Distributions
11	d	Random Variables	25	e	Continuous Distributions
12	c	Random Variables	26	d	Continuous Distributions
13	a	Discrete Distributions	27	b	Continuous Distributions
14	c	Discrete Distributions	28	d	Continuous Distributions